

An objective approach for feature extraction:

distribution analysis and statistical descriptors for scale choice and channel network
identification

GES 419

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Purpose: Research Gap?

Traditional extraction methods using DTMs:

- Use fixed thresholds (area or slope-area ratio) that may not apply across different terrains.
- Are sensitive to data quality and morphology.
- Often overpredict channel heads, especially in complex terrains.



Objective #1

- Select the scale for terrain analysis w/o bias.
- Automatically identify channel networks using LiDAR sourced topographic attributes.

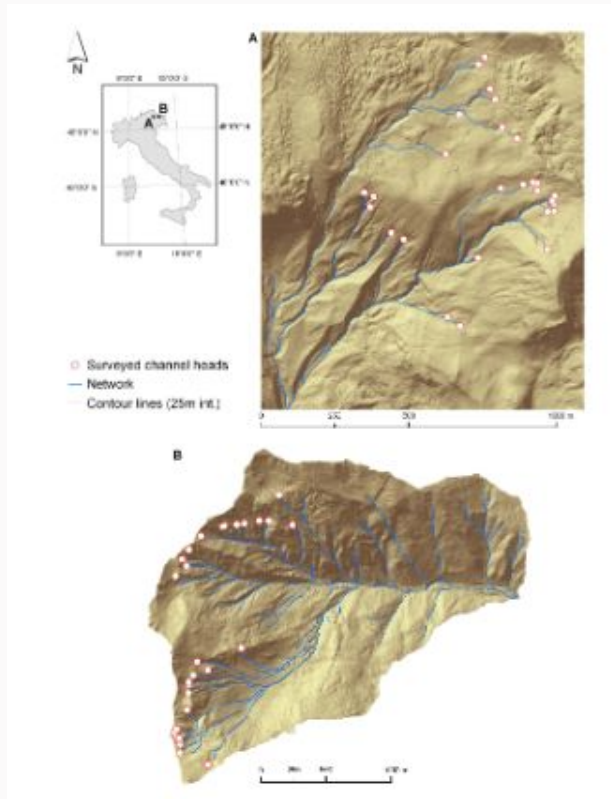


~~Objective #1~~ Avoid subjectivity (no set manual threshold).

- Be applicable to varied complex landscapes

Study Area

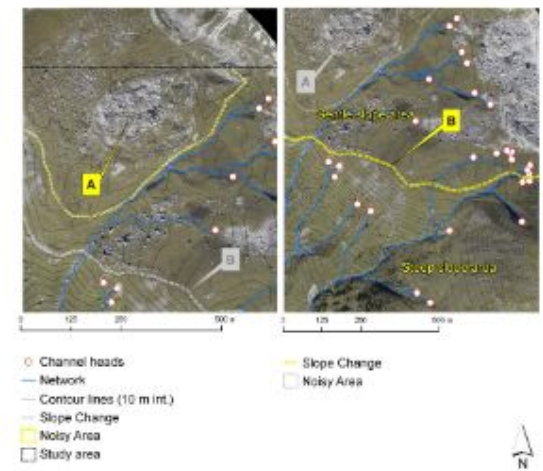
Cordon Basin 1.4 km²



Miozza Basin 4.4 km²

Common Attributes

- High-resolution 1m LiDAR DTM.
- Field-mapped channel networks and heads (DGPS).
- Wide range of contributing areas for channel heads → generic thresholds are ineffective.



Cordon has a simple shape but morphologically complex (abrupt slope changes)

Miozza is a more organized catchment but has complex terrain.

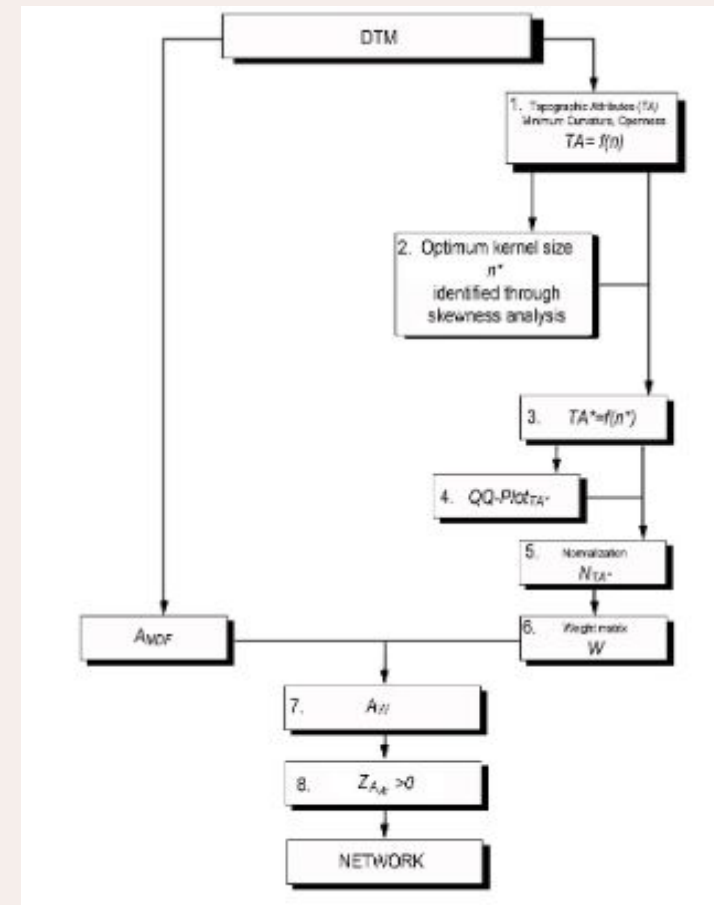
Step 1

Morphology Detection Using Topographic Attributes

Two terrain metrics:

1. Minimum Curvature (Cmin)
 - a. Measures surface concavity and highlights valleys or channels.
2. Openness (+ & -)
 - a. Measures how open or enclosed a point is; captures general shape trends.

Methodology



Step 2

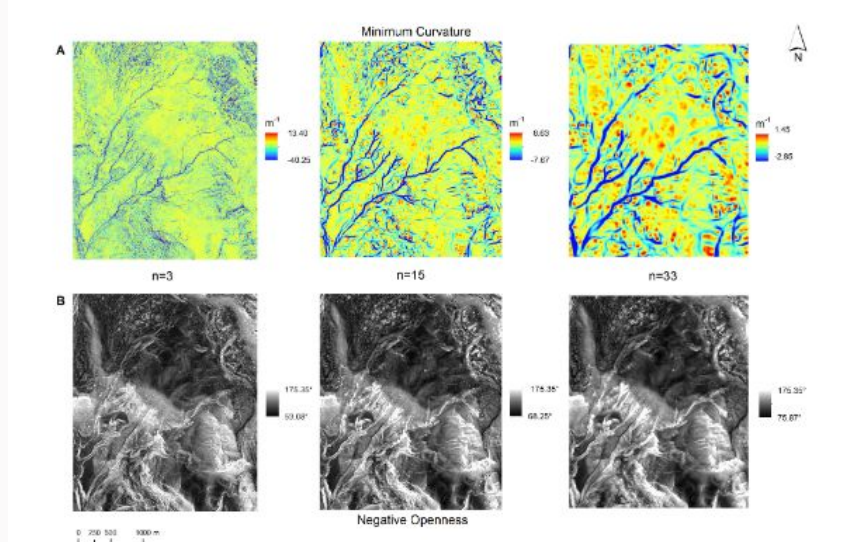
Kernel Size Selection

Problem:

The effect of window size changes the accuracy of curvature and openness detection.

Solution:

They analyze the skewness of the value distribution for each kernel size:



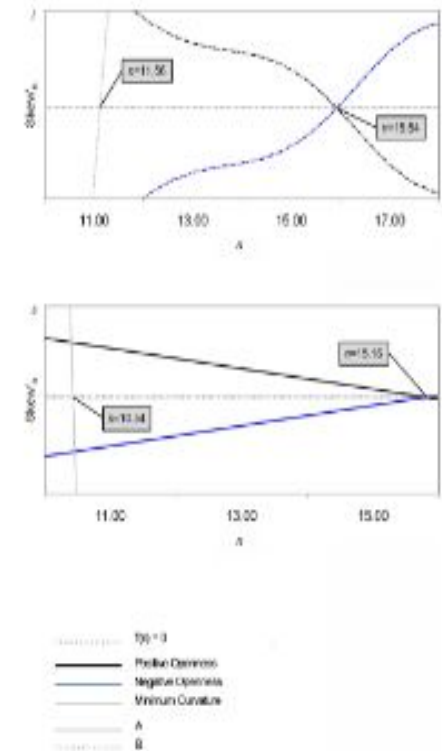
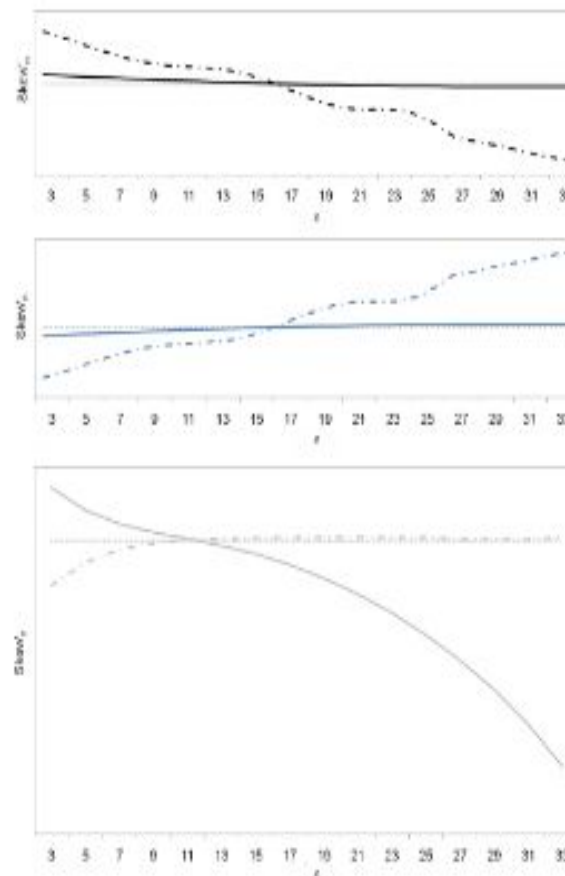
- If values become more skewed (asymmetric), it means the kernel is enhancing features like valleys for example.
- When skewness stops increasing, the kernel is too large = smoothing out important details.

Step 3

Normalization and Weighting

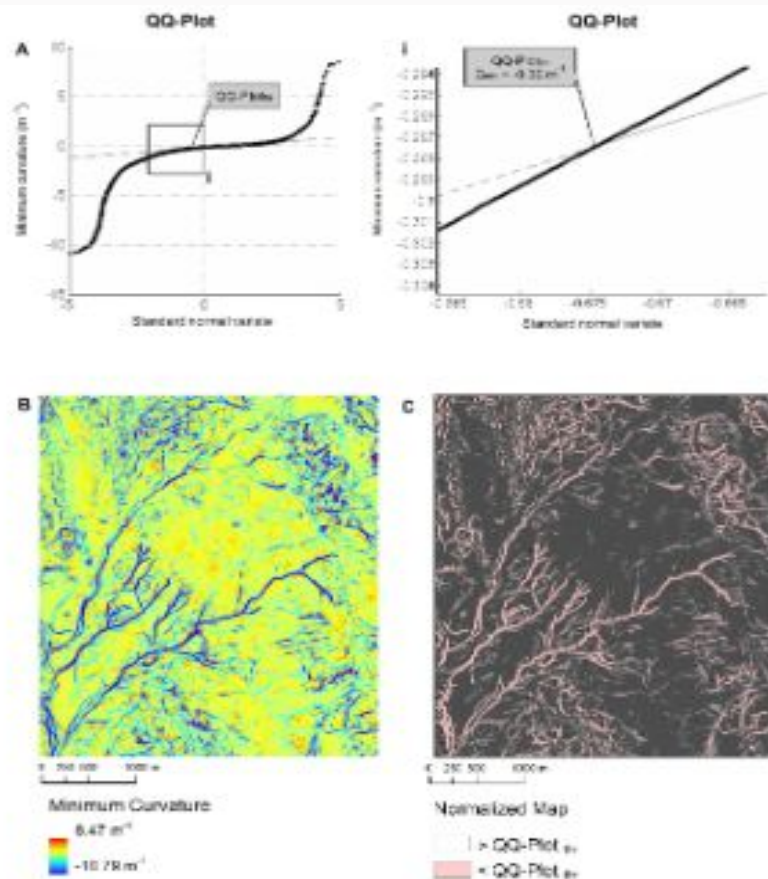
Combine terrain attributes & reduce artifacts:

1. Normalize each map using QQ-plot thresholds.
 - Areas of extreme curvature or openness are significant.
2. Combine them into a weight matrix (W):
 - $W = N(Cmin) \cdot N(\psi L) \cdot (1/N(\phi L))$
 - i. ψL = negative openness, ϕL = positive openness.



Step 4

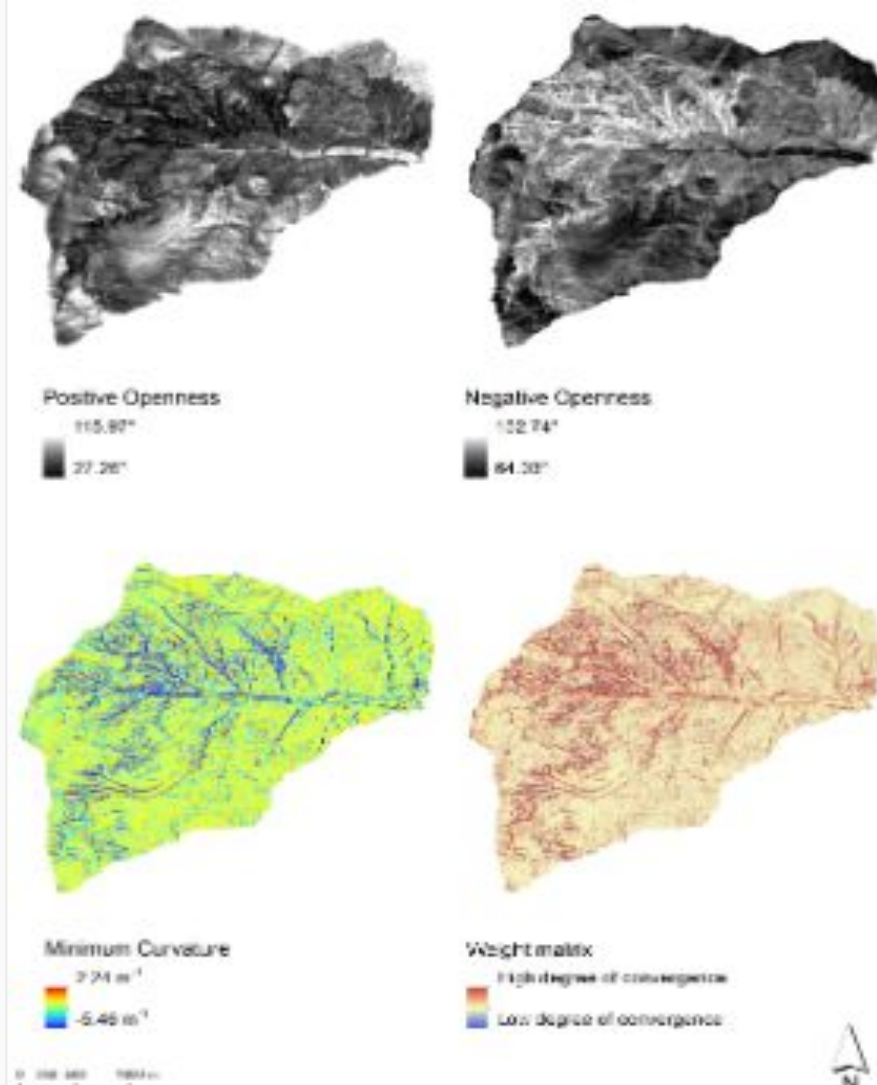
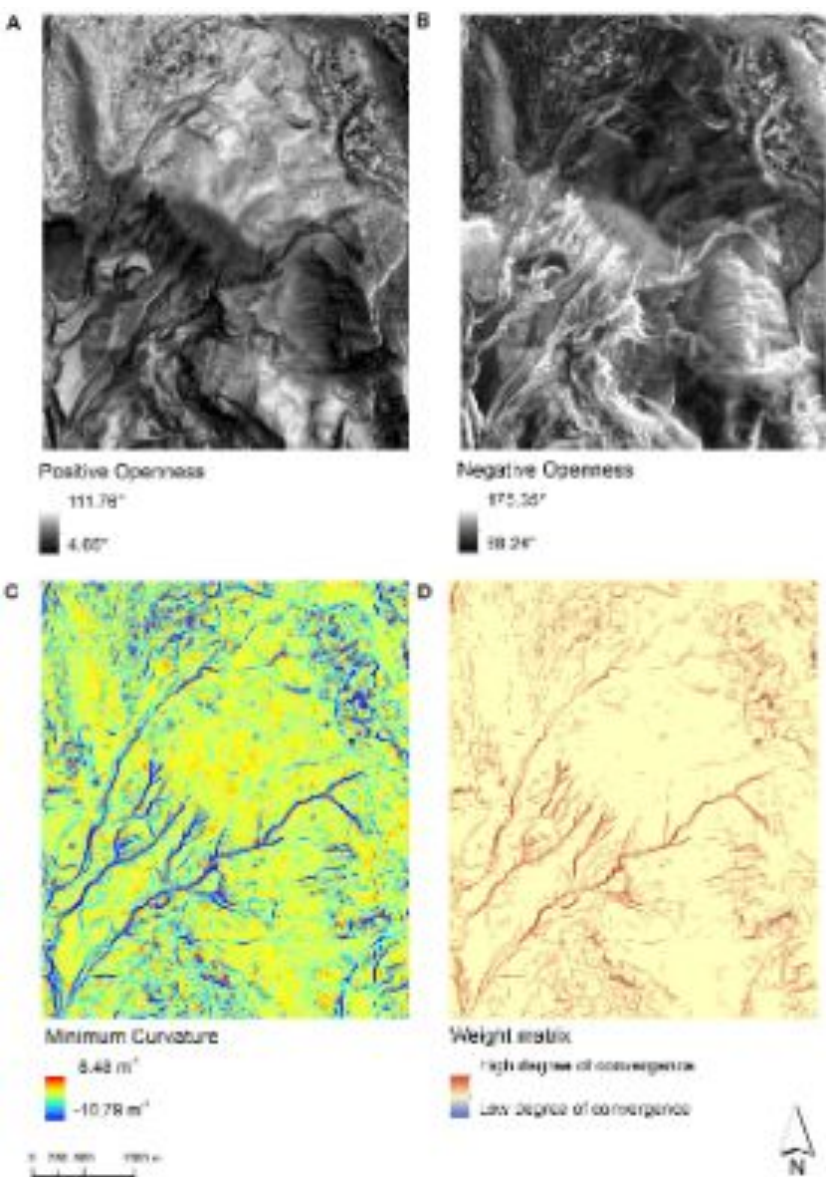
Flow Convergence



Modified Multiple Flow Direction algorithm (Quinn et al., 1991):

Instead of distributing flow only based on slope, they **weight** it using the morphology-based W matrix.

Simulates realistic drainage paths where convergence is high.



Step 6

Noise Filtering and Network Connection

Miozza:

- Simple majority filter removes noisy isolated pixels.

Cordon):

- Use local entropy: measures randomness in flow directions.
- High entropy → noise; low entropy → structured channel.
Noisy segments above average are removed.

Final Step: Connect all channels to the main pour point using a least-cost path algorithm over Euclidean distances.